STAFF REVIEW · H446 244 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. The address of the memory location being accessed.

2. Only when both operands are true.

3. Incremental working versions and repeated feedback.

4. low=0 high=14 → mid=7; target greater → low=8 high=14 mid=11; greater → low=12 high=14 mid=13; found. Midpoints 7,11,13.

I Do — source worked steps

1. Predict the method — Add all durations for the sequential path: 10 minutes.

2. Check and explain — Run independent checks together, wait 5 minutes, then publish in 2: total 7 minutes.

We Do — guided CHECK

1. Predict the method
Images and CSS are independent preparation tasks; calculate 4+6+1=11 minutes sequentially.
Check and explain
Run both together; CSS sets the critical 6-minute time. Then deploy 1 minute: 7 minutes.

You Do — source mark schemes

1. [4] Sequential 3+5+2=10 minutes [2]. Parallel max(3,5)+2=7 minutes [2].

2. [6] Both read £100 [1]; separate writes of £90 and £80 can leave £80 or £90 instead of £70 [2]. A lock around the whole read-modify-write, or a suitable atomic transaction, serialises the updates [2]; waiting can reduce concurrency [1].

3. [6] A and B overlap [1]; complete after max(4,6)=6 [1]; C then 2 gives 8 [1]; D then 1 gives 9 minutes [1]; overhead [1] and limited shared resources/communication [1].

Exit Ticket — source answer

1. False. Dependencies and shared-state constraints may require sequential execution.